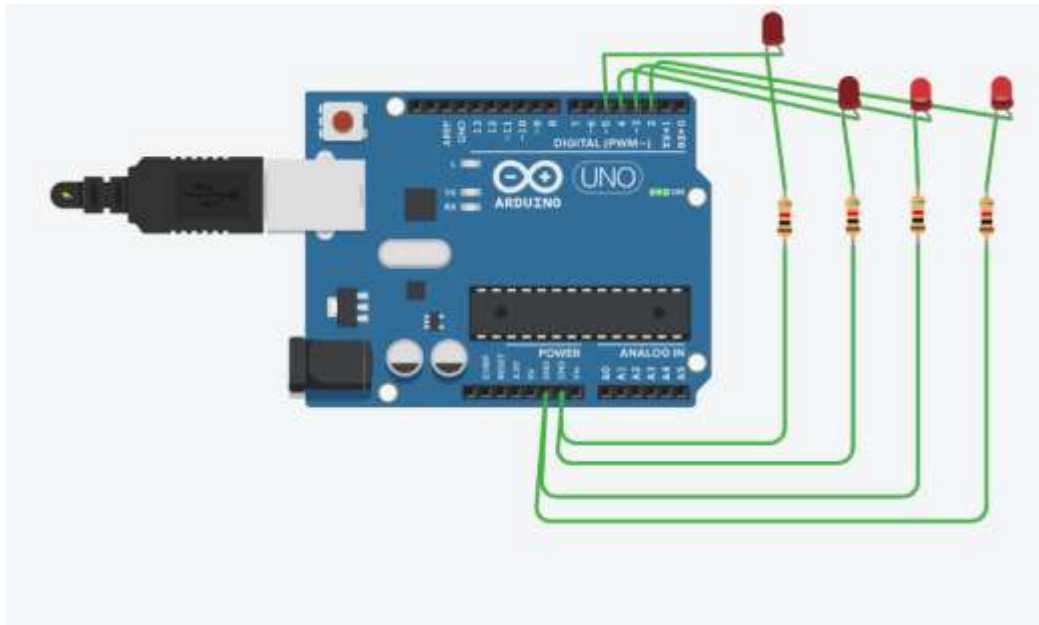




Text

1 (Arduino Uno R3)

```
1 int LED1 = 5;
2 int LED2 = 4;
3 int LED3 = 3;
4 int LED4 = 2;
5 void setup() {
6   Serial.begin(9600);
7   pinMode(LED1, OUTPUT);
8   pinMode(LED2, OUTPUT);
9   pinMode(LED3, OUTPUT);
10  pinMode(LED4, OUTPUT);
11  Serial.println("Enter a decimal number (0 to 15):");
12 }
13 void loop() {
14   if (Serial.available() > 0) {
15     int decimalVal = Serial.parseInt();
16     if (decimalVal >= 0 && decimalVal <= 15) {
17       Serial.print("Decimal: ");
18       Serial.print(decimalVal);
19       Serial.print(" -> Binary: ");
20       int bit3 = (decimalVal >> 3) & 1;
21       int bit2 = (decimalVal >> 2) & 1;
22       int bit1 = (decimalVal >> 1) & 1;
23       int bit0 = (decimalVal >> 0) & 1;
24       Serial.print(bit3);
25       Serial.print(bit2);
26       Serial.print(bit1);
27       Serial.println(bit0);
28       digitalWrite(LED1, bit3 ? HIGH : LOW);
29       digitalWrite(LED2, bit2 ? HIGH : LOW);
30       digitalWrite(LED3, bit1 ? HIGH : LOW);
31       digitalWrite(LED4, bit0 ? HIGH : LOW);
32     } else {
33       Serial.println("Error: Please enter a valid number between 0
34     }
35     while (Serial.available() > 0) {
36       Serial.read();
37     }
38   }
39 }
```